

TryHackMe-DNS in Detail

1. Introduction

DNS (Domain Name System) is used to translate human-readable domain names into IP addresses. Instead of remembering a numerical IP address, users can use an easy-to-remember domain name such as tryhackme.com. DNS helps the computer find the correct IP address and connect to the required server.

2. Domain Hierarchy

A domain name consists of different levels.

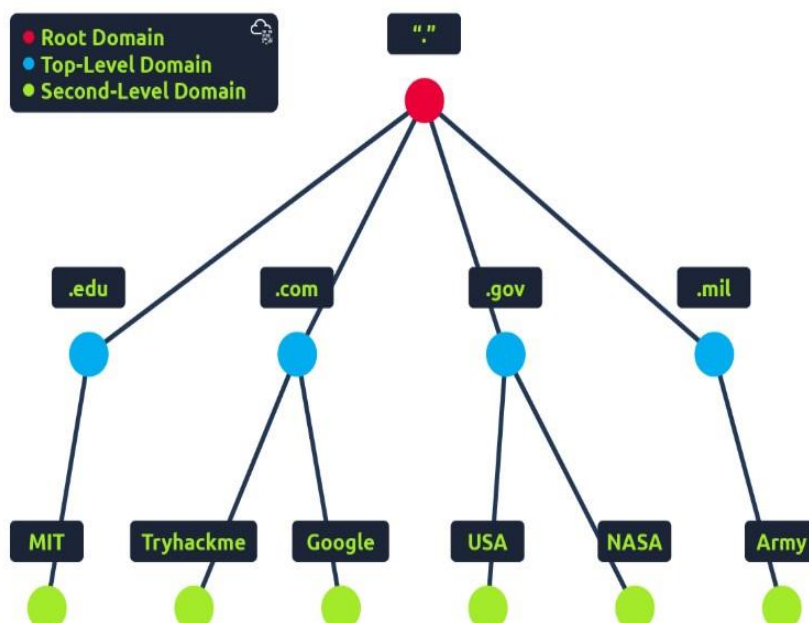
For example, in:

admin.tryhackme.com

- **.com** – Top-Level Domain (TLD)
- **tryhackme** – Second-Level Domain
- **admin** – Subdomain

TLDs can be generic, such as .com, .org, and .edu, or country-based, such as .in, .uk, and .ca.

Domain Hierarchy



3. DNS Record Types

DNS uses different records to store different types of information.

Record Purpose

A Maps a domain to an IPv4 address

AAAA Maps a domain to an IPv6 address

CNAME Points a domain to another domain

MX Specifies mail servers

TXT Stores text-based information

DNS Record Types

DNS isn't just for websites though, and multiple types of DNS record exist. We'll go over some of the most common ones that you're likely to come across.

A Record

These records resolve to IPv4 addresses, for example 104.26.10.229

AAAA Record

These records resolve to IPv6 addresses, for example 2606:4700:20::681a:be5

CNAME Record

These records resolve to another domain name, for example, TryHackMe's online shop has the subdomain name store.tryhackme.com which returns a CNAME record shops.shopify.com. Another DNS request would then be made to shops.shopify.com to work out the IP address.

MX Record

These records resolve to the address of the servers that handle the email for the domain you are querying, for example an MX record response for tryhackme.com would look something like alt1.aspmx.l.google.com. These records also come with a priority flag. This tells the client in which order to try the servers, this is perfect for if the main server goes down and email needs to be sent to a backup server.

TXT Record

TXT records are free text fields where any text-based data can be stored. TXT records have multiple uses, but some common ones can be to list servers that have the authority to send an email on behalf of the domain (this can help in the battle against spam and spoofed email). They can also be used to verify ownership of the domain name when signing up for third party services. Here are a few examples:

- `_acme-challenge.example.com TXT "token_value_here"`
- `@ TXT "v=spf1 ip4:192.0.2.0/24 include:_spf.google.com include:amazonses.com ~all"`

4. DNS Request Process

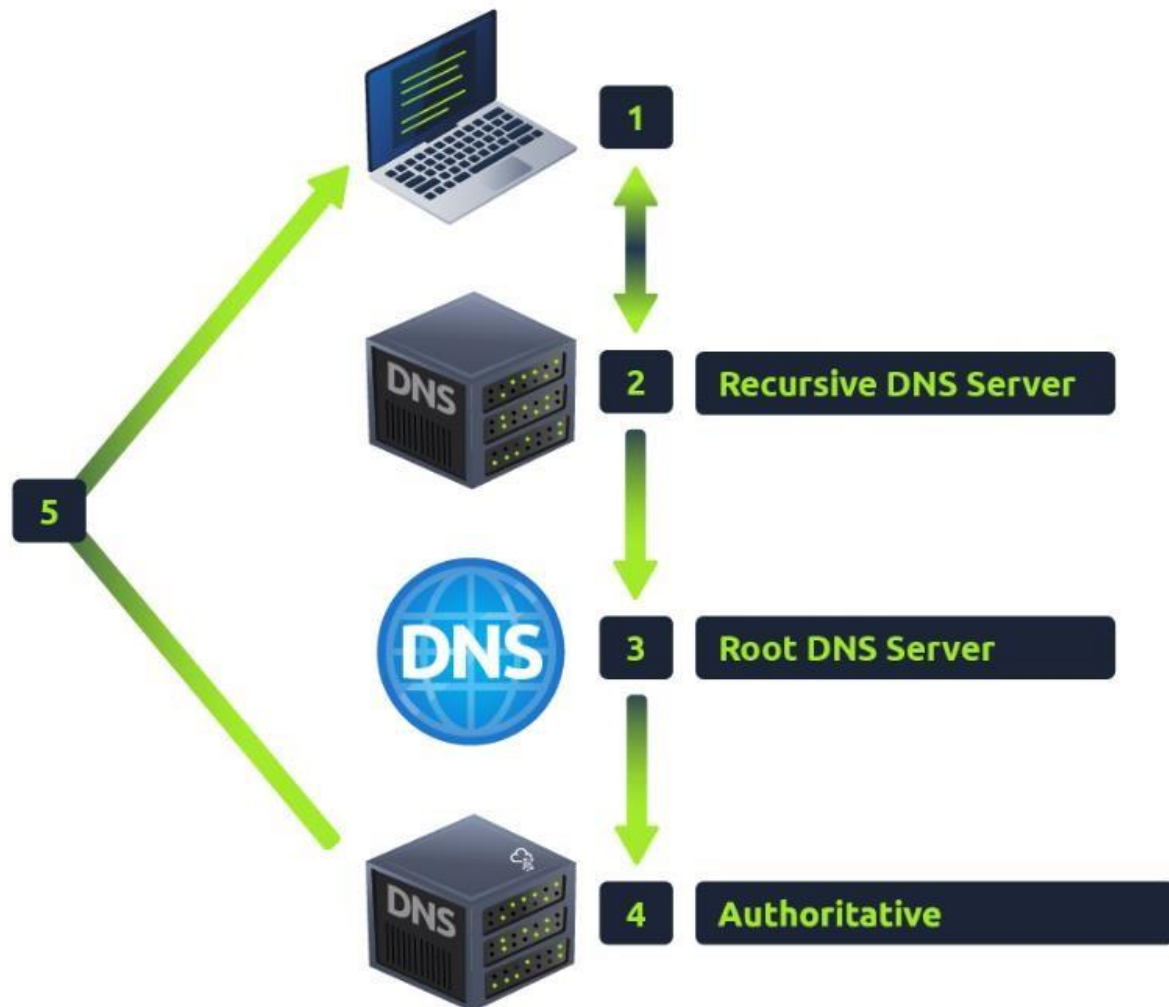
When a user requests a domain, the computer first checks its local DNS cache. If the information is not available, the request goes to a **Recursive DNS Server**.

The recursive server may then contact:

Root Server → TLD Server → Authoritative DNS Server

The authoritative server provides the required DNS record. The result is sent back to the user and can be cached for future requests.

DNS records also have a **TTL (Time To Live)** value, which determines how long the result can remain cached.

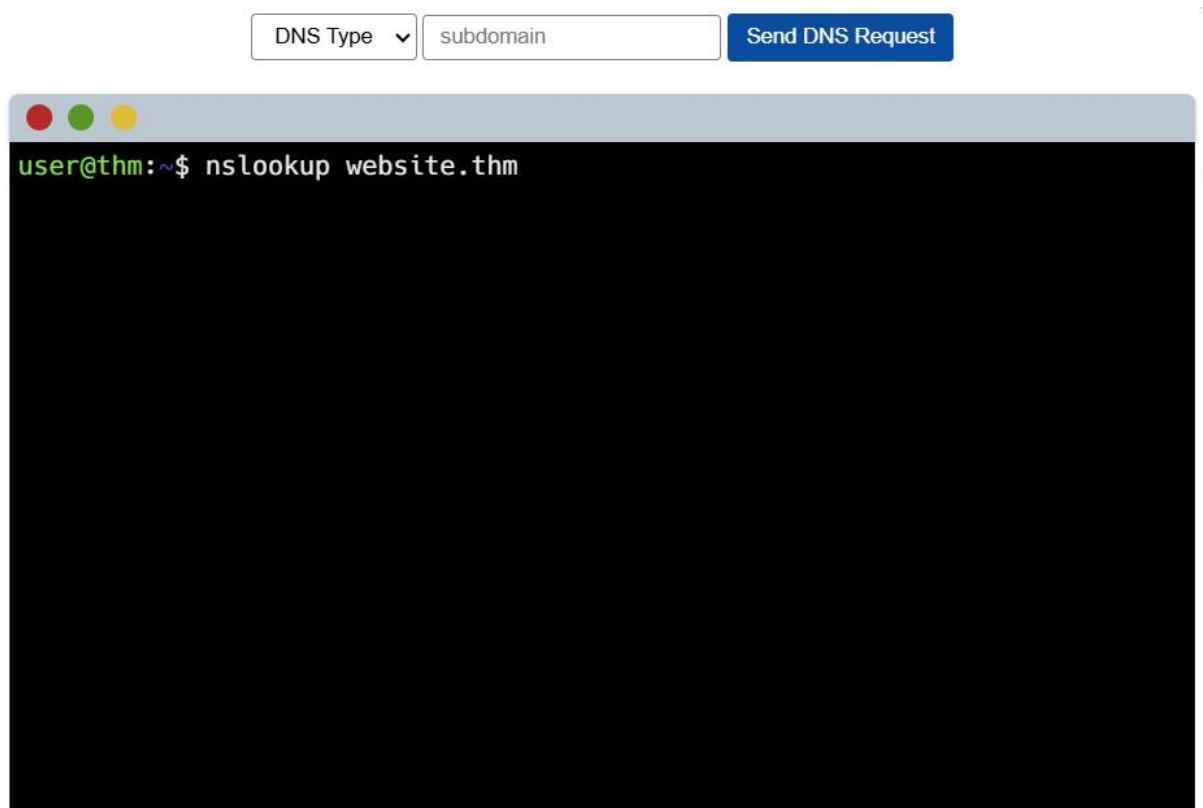


5. Practical DNS Queries

The TryHackMe DNS tool was used to query different DNS records.

Queries Performed

1. **CNAME:** shop.website.thm
2. **TXT:** website.thm
3. **MX Priority:** website.thm
4. **A Record:** www.website.thm



6. TryHackMe Questions and Answers

Question

Answer

What does DNS stand for?

Domain Name System

What is the maximum length of a

63

subdomain?

Which of the following characters cannot

-

be used in a subdomain (3 b _ -)?

What is the maximum length of a domain

253

name?

What type of TLD is .co.uk?

ccTLD

What type of record would be used to

MX

advise where to send email?

What type of record handles IPv6

AAAA

addresses?

What field specifies how long a DNS record

TTL

should be cached for?

Question

Answer

What type of DNS Server is usually **recursive** provided by your ISP?

What type of server holds all the records **authoritative** for a domain?

What is the CNAME of shop.website.thm? **shops.myshopify.com**

What is the value of the TXT record of

THM{7012BBA60997F35A9516C2E16D2944FF}

website.thm?

What is the numerical priority value for the

30 MX

record?

What is the IP address for the A record of

10.10.10.10

www.website.thm?

7. Learning Outcomes

After completing this room, I learned:

- How DNS translates domain names into IP addresses.
- Different levels of domain hierarchy.
- Common DNS record types.
- How DNS resolution works.
- The role of recursive, root, TLD, and authoritative servers.
- The purpose of DNS caching and TTL.
- How to perform basic DNS queries.

8. Conclusion

This TryHackMe room provided a basic understanding of DNS and its role in networking. The practical queries helped me understand how different DNS records are used and how DNS servers work together to resolve domain names.

https://tryhackme.com/room/dnsindetail?utm_campaign=social_share&utm_medium=social&utm_content=share-completedroom&utm_source=copy&sharerId=65814d9d59374f0cfeab042c

Harshika Bhosale